

Exercise 04

PART A

For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet.

1. In the database application system lifecycle, _____.
 - A. database design must be done before application design
 - B. application design must be done before database design
 - C. database design should be done concurrently with application design
 - D. application design must be done after logical database design
2. In the database application system lifecycle, two different user views of a database application _____.
 - A. must contain common data
 - B. may contain common data or not
 - C. never contain common data
 - D. must contain parts of each other's data
3. An entity type that is not existence-dependent on some other entity type is called _____.
 - A. a strong entity type
 - B. a weak entity type
 - C. an independent entity type
 - D. an abstract entity type
4. An entity type that is existence-dependent on some other entity type is called _____.
 - A. a strong entity type
 - B. a weak entity type
 - C. a dependent entity type
 - D. a reduced entity type
5. [☆] The meaning of the only '0' in the following ER diagram means that _____.
 - A. it is an one-to-one relationship
 - B. it is an one-to-zero relationship
 - C. a member of staff is not necessarily a branch manager
 - D. there may exist a branch that has no manager



6. [☆] The meaning of the only '*' in the following ER diagram means that _____.
 - A. a member of staff may oversee multiple properties for rent
 - B. there may exist many properties for rent that are not overseen by any member of staff
 - C. it is an one-to-many relationship
 - D. it is a zero-to-many relationship



7. [☆] The meaning of the only '*' in the following ER diagram means that _____.
 - A. there may exist many clients that are not registered by any member of staff at any branch
 - B. a member of staff may register multiple clients at a branch
 - C. it is an one-to-many relationship
 - D. it is an one-to-one-to-many relationship



8. [☆] The meaning of the only '0' in the following ER diagram means that _____.
 A. there may exist a client that is not registered by any member of staff at any branch
 B. it is an one-to-many relationship
 C. a member of staff may not register any client at a branch
 D. it is an one-to-zero-to-many relationship



9. [☆] A process that maximizes differences between members of an entity type by identifying their distinguishing characteristics is called a _____.
 A. maximization process
 B. generalization process
 C. specialization process
 D. distinguishing process
10. [☆] A process that minimizes differences between entities by identifying their common characteristics is called a _____.
 A. minimization process
 B. generalization process
 C. specialization process
 D. homologizing process
11. In deriving relations from an ER diagram, the correct approach to process a 1:* binary relationship type is _____.
 A. copying primary key of '1' side into relation of '*' side to act as a foreign key
 B. copying primary key of '*' side into relation of '1' side to act as a foreign key
 C. copying a foreign key of '1' side into relation of '*' side to act as primary key
 D. copying a foreign key of '*' side into relation of '1' side to act as primary key
12. In deriving relations from an ER diagram, the correct approach to process a *:~ binary relationship type is _____.
 A. copying primary key of one side into relation of the other side to act as a foreign key
 B. copying a foreign key of one side into relation of the other side to act as primary key
 C. representing the relationship using a relation, copying primary keys of both sides into it
 D. adding an attribute into each side of the relationship to express the rank information
13. [☆] Suppose we are given a ER diagram that is composed of 3 entity types and 3 *:~ binary relationship types, then we can derive ____ relations from this ER diagram.
 A. 4
 B. 5
 C. 6
 D. 7
14. In deriving relations from an ER diagram, which one of the following is wrong?
 A. Creating a relation for each entity type
 B. Creating a relation for each relationship type
 C. Creating a relation for each *:~ binary relationship type
 D. Generally it is unnecessary to create relations for 1:1 and 1:* binary relationship types
15. [☆] When merging local logical data models into a global model, 'homonym' [同名异义] means that _____.
 A. two or more entities/relations have the same name but are different

- B. two or more entities/relations are the same but have different names
 - C. two or more entities/relations are derived from the same concepts
 - D. two or more entities/relations are the same but derived from different concepts
16. [☆] When merging local logical data models into a global model, 'synonyms' [异名同义] means that ____.
- A. two or more entities/relations have the same name but are different
 - B. two or more entities/relations are the same but have different names
 - C. two or more entities/relations are derived from the same concepts
 - D. two or more entities/relations are the same but derived from different concepts

PART B

Fill in blanks with proper words or phrases.

1. A database application system lifecycle include 10 steps, i.e.: (a), (b), (c), (d), (e), (f), (g), (h), (i), (j).
2. Database design consists of 3 phases: (a) database design, (b) database design, (c) database design.
3. [☆] An entity type is a group of (a) with the same (b), which can be represented in ER diagram using a (c) labeled with entity name; a relationship type is a set of (d) among (e); an attribute is a property of an entity type or a (f).
4. In deriving relations from an ER diagram, the correct approach to process a 1:* binary relationship type is copying (a) of '1' side into relation of '*' side to act as a (b), the correct approach to process a *: * binary relationship type is representing the relationship using (c), copying (d) of both sides into it.
5. An entity is a (a) that is to be represented in the database; an attribute is a (b) of an entity; a relationship is an (c) between entities.

PART C

Database Design.

In a library, a reader can borrow zero or more books; a book can be borrowed by zero or more readers; an author can write one or more books, but for each book, we only store its first author. Books should have attributes such as book number, book name, publisher, price; readers should have attributes such as reader number, reader name, reader gender, reader address; authors should have attributes such as author number, author name, author gender, author affiliation. When a reader borrows a book, we must register the borrowing date.

- (1) Draw corresponding ER diagram.
- (2) Derive relation data model from your ER diagram.
- (3) Give the SQL statements to create table in your data model such that :
 - a) primary key and foreign keys of each table are properly defined;
 - b) book name, author name, reader name, author gender and reader gender cannot be null;
 - c) gender can only take value 'M' or 'F'.